

Assignment-1 CA-Software

MVC Design Pattern Explanation:

The MVC design pattern is a software architectural pattern used to separate an application into three main components: Model, View & Controller. The separation helps organize code for better maintainability, scalability & testing.

1) Model:

- 1) Represents the data & business logic.
- 2) Manages data, responds to requests for information, & updates the view when the data changes.

2) View:

- 1) Represents the user interface.
- 2) Displays data from the model & sends the user input to Controller.

3) Controller:

- 1) Acts as an intermediary between the model & view.
- 2) Takes input from the user, processes it & updates the view.

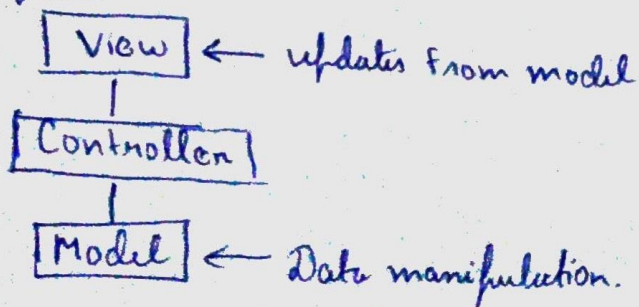
MVC Variants:

- 1) Classic MVC: In this variant, the view directly interacts with the model. The controller processes input, updates the model & notifies the view of changes.

When to use:

- Ideal for applications where you need strict separation between the UI & business logic, such as traditional web apps.

Diagram:



23/ Model View Presenter (MVP):-

Here the presenter replaces the controller & takes over the responsibility of updating both the Model & the view. The view is passive & relies on the presenter for all interactions.

When to use:

Best for applications with complex UI interactions, like desktop or mobile apps, where the view is decoupled from the logic.

Diagram:

